EcoLoop: Presentation Speaking Pointers

Quick reference cheat sheet for the 15-slide deck.

Swami Sant Dass Public School
github.com/Harman209/ecoloop

SLIDE 01

Title: EcoLoop

- Frame as an engineering system, not a craft or poster project.
- Two distinct halves: physical materials + $0 digital infrastructure.
- Affiliation: Swami Sant Dass Public School, Jalandhar.

SLIDE 02

The Linear Packaging Problem

- Multi-layer packaging (MLP): chips wrappers bonding plastic & foil.
- Municipal recyclers dump 100% of it because separation is unprofitable.
- Our fix: heat-press MLP into structural boards replacing plywood.

SLIDE 03

The Physical Loop (6 Materials)

- 6 verified items: Eco-Bricks, MLP Sheets, Bottle Planters, Uniform Bags, Seed Paper, Bioenzymes.
- Point to photo: real heat-pressed sample board made of snack wrappers.
- Durable, waterproof, termite-proof, drills and saws like timber.

SLIDE 04

Reality Check 01: Transition

- Joke: teachers thought we would just put colored dustbins in school.
- Reality: managing specs and orders required cloud software in 2 weeks.
- Transition: moving from picking up wrappers to building software.

SLIDE 05

Why Projects Need Software

- Without engineering specs, nobody takes recycled materials seriously.
- Platform publishes density, dimensions, thermal ratings, lead times.
- Built with React 18, TypeScript, and Vite for fast client routing.

SLIDE 06

The Zero-Cost Serverless Backend

- Core constraint: zero budget, no corporate credit cards.
- Pipeline: React form → Google Apps Script webhook → Google Sheets.
- Result: $0.00/mo operating cost, zero maintenance, 99.9% uptime.

SLIDE 07

Reality Check 02: Transition

- Enterprise devs spend $450/month debating Kubernetes and Docker.
- High schoolers run production orders on Google Sheets and it never crashes.

SLIDE 08

Interactive Inquiry Engine

- Selecting 'Enquire' pre-populates product ID, specs, and quantities.
- Client-side validation catches mistakes before sending data.
- Instant confirmation modal + direct logging into backend sheet.

SLIDE 09

Developer as Architect

- Old way: typing repetitive boilerplate, fighting syntax, 3 hours debugging.
- Modern way: developer designs constraints & types; AI agents write code.
- Human role shifts to architecture, boundaries, and verification.

SLIDE 10

Reality Check 03: Transition

- Legacy advice: read 400 pages of framework documentation first.
- Reading manuals at 3:00 AM is dead; write clean specs and let agents build.

SLIDE 11

The Autonomous Coding Loop

- 1. Define goal: clear spec document with design rules & types.
- 2. Autonomous edit: agent reads codebase and modifies files.
- 3. Automated check: compiler runs, catches errors, agent self-repairs.
- 4. Safe history: git checkpoints let you revert experiments instantly.

SLIDE 12

Introducing OpenCode

- 100% open-source terminal agent (GitHub: anomalyco/opencode).
- Plan mode (surveys codebase safely) vs Build mode (executes file edits).
- One command install: `npm install -g @opencode/cli`.

SLIDE 13

Free AI Models for Students

- Free cloud: Google Gemini (AI Studio), Groq Cloud, GitHub Models.
- 100% offline: Ollama running `qwen2.5-coder:7b` without internet.
- Zero credit cards, zero subscriptions.

SLIDE 14

Reality Check 04: Transition

- AI model claims: "I fixed all the bugs, looks perfect!"
- Compiler reality: 47 errors found in 3 files.
- Rule of thumb: never trust claims; always verify with the compiler.

SLIDE 15

Thank You & Scannable QR

- Scan QR code: opens GitHub repository with full source code and live demo.
- Take-home: hand out the 1-page setup guide so students can install tools tonight.
- Open the floor for questions from students and teachers.